

Indian Statistical Institute
Second Semester Examinations: 2025-26

Course Name: M. Math, 2nd year
Subject Name : Probability Theory
Maximum Marks: 25, Duration: Two hours
Date: 13.09.2025, 11:00 AM – 1:00 PM

- You may use any results proved in class. Any other results require proof.
- Between Question 2(c) and Question 4(c), your answers will be scored for a total of 5 marks.
- You will be awarded the full marks for the question if you justify using correct arguments that there is a mistake in the question

1. (a) [2.5 marks] Construct a probability space $(\Omega = \{0, 1\}^2, \mathcal{F}, \mathbb{P})$ and random variables $X : \Omega \rightarrow \{0, 1\}$ and $Y : \Omega \rightarrow \{0, 1\}$ which are identically distributed but fail to be independent. Assume here that $\{0, 1\}$ is equipped with the power-set σ -algebra.
- (b) [2.5 marks] With data/notation as in part (a), construct RVs X and Y which are independent but not identical.
2. (a) [2 marks] Let $(X_i)_{i=1}^\infty$ be an i.i.d sequence of RVs which models the rolling of a fair die, that is, $\mathbb{P}[\{X_i = j\}] = 1/6$, for each $i \in \mathbb{N}, 1 \leq j \leq 6$, and assume that the die-rolls are independent. Compute the probability of getting "1" within the first n die-rolls in terms of n .
- (b) [1 marks] Let $\tau(\omega) = \min\{i \in \mathbb{N} \mid X_i(\omega) \neq 1\}$ for any ω in the probability space. Compute $\mathbb{P}[\{\omega \mid \tau(\omega) = 6\}]$.
- (c) [5 marks] Define inductively for $k \in \mathbb{N}$,

$$\tau_{k+1}(\omega) = \min\{i > \tau_k(\omega) \mid X_i(\omega) \neq 1\},$$

where $\tau_1 = \tau$. Define

$$S_{\tau_k}(\omega) := \sum_{i=1}^{\tau_k(\omega)} X_i(\omega).$$

Determine

$$\lim_{i \rightarrow \infty} \frac{S_{\tau_k}}{\tau_k}$$

for almost every sequence of infinite die-rolls, if it exists.

3. Let $(X_i)_{i=1}^\infty, (Y_i)_{i=1}^\infty, X, Y$ be $\mathbb{R}$ -valued random variables in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, such that $X_i \rightarrow X$ and $Y_i \rightarrow Y$ in $\mathbb{P}$, that is, in probability, as $i \rightarrow \infty$.
- (a) [3 marks] Show that $X_i Y_i \rightarrow XY$ in $\mathbb{P}$ if all the RVs X_i, Y_i, X, Y are $\mathbb{P}$ -almost surely bounded in absolute value by $0 < M < \infty$. Here XY means the RV defined $XY(\omega) = X(\omega)Y(\omega)$.
- (b) [2 marks] Prove the assertion in part (a) without the assumption of boundedness.
- (c) [3 marks] Show that if

$$\sum_{i=1}^{\infty} \mathbb{P}[\{\omega \in \Omega \mid |X_i(\omega) - X(\omega)| > \epsilon_i\}] < \infty,$$

for a sequence $(\epsilon_i)_{i=1}^\infty$ where $\epsilon_i \rightarrow 0$ then $\lim_{i \rightarrow \infty} X_i = X$ almost surely.

4. Let $(X_i)_{i=1}^{\infty}$ be an independent sequence of RVs, with values in $[0, \infty)$. Define for any $\alpha \geq 0$, $S_k^{(\alpha)} = \sum_{i=1}^k X_i^\alpha$ for $k \in \mathbb{N}$.

- (a) [2 marks] Suppose X_1 has distribution $\mathbb{P}[\{X_1 \in [0, a]\}] = \int_0^a e^{-x} dx$. Verify that for any $\alpha > 0$, $\epsilon > 0$,

$$\frac{S_n^{(\alpha)}}{n^{1+\epsilon}} \xrightarrow{n \rightarrow \infty} 0.$$

- (b) [2 marks] Prove the converse of Marcinkiewicz-Zygmund: if $\mathbb{E}[X_1^2] = \infty$, but $\mathbb{E}[X_1] = 1$ and for some $1 < p < 2$, $\frac{S_n^{(1)} - n}{n^{1/p}}$ converges almost surely, then show that

$$\mathbb{E}[X_1^p] < \infty.$$

- (c) [5 marks] Let $(X_i)_{i=1}^{\infty}$ model the independent tossing of a fair coin and assume that you participate in the following scheme based on outcomes of coin tosses. You start with a certain balance in rupees. Then at the k -th toss, when k is odd, you win five rupees if a heads appears and you lose four rupees if a tails appears. When k is even however, you win two rupees if a heads appears and lose one rupee if a tails appears. Suppose that you are only allowed to continue this exchange beyond the k -th toss if your total balance does not sum up to zero at the k -th toss. Show that the probability that you have to quit due to zero balance after the m -th toss, if you started with an initial balance of m rupees is not more than $90/m$.